



T4 CAPSTONE: BECOME A JUNIOR INVENTOR!

Design Your Own Circuit to
Solve Real Problems

YOUR MISSION AS JUNIOR INVENTORS!

The miniature toy village is pitch dark at night!
The villagers need a street light that can turn
ON and OFF. As Inventors, your goal is to
design a functional circuit to light up the
village!

Complete Loop – Electricity must travel in
a full, unbroken path!

Bridge Control – A switch opens or
closes the circuit path.



CONCEPT A: THE CLOSED HIGHWAY!

Electricity travels like a smooth, unbroken highway — from the battery, through the wires, and back again!
Just like a toy train's railway track (metro line) — if one section of the track is missing, the train stops completely. No gap = smooth journey!
Trace the path and make sure it's a full loop!

Complete Loop = Electricity flows!
A gap = no power!



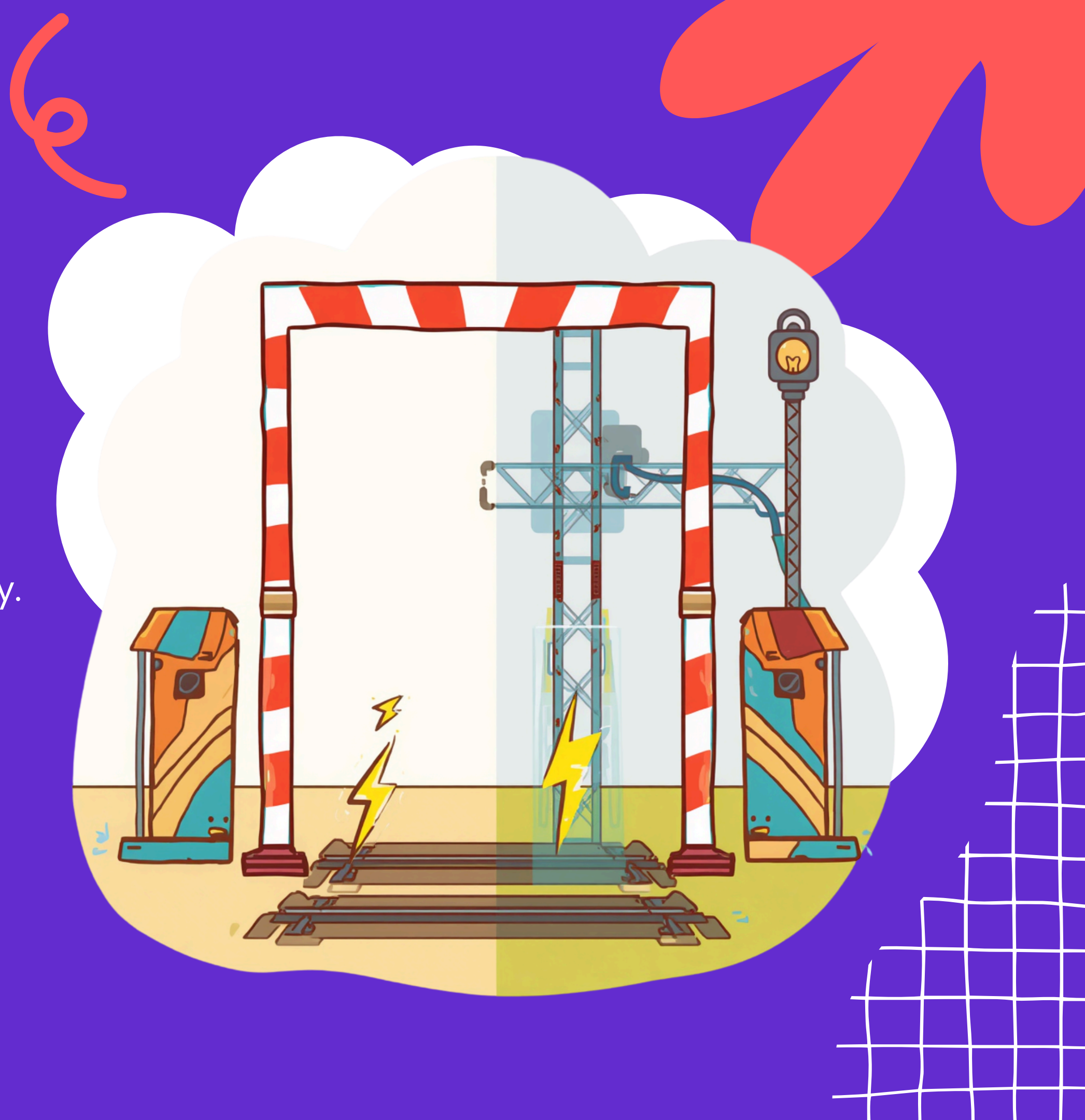
CONCEPT B: THE GATEKEEPER!

The Switch Controls Everything

The switch is like a railway crossing gate (phaatak) that controls the flow of electricity. When the gate is DOWN, electricity flows freely. When the gate is UP, electricity stops completely!



Bridge Control: You decide when electricity travels!



CONCEPT C: BUILD IT!

Combine all four parts to solve a real problem!
Energy giver (battery) + energy highway (wire)
+ gatekeeper (switch) + shining star (LED) = a
working circuit!

Battery gives power
to the whole circuit.

Switch controls
when the light
turns on or off.

Wires carry
electricity like a
highway.

LED lights up a
Gharunda — just like
Diwali!



THE MASTER BLUEPRINT PUZZLE!

Can you connect four parts without crossing lines or dead ends? Use your inventor skills to complete the circuit blueprint and light up the village!

Parts: battery, push-button switch, wires, and small buzzer.

Click and connect the blocks to complete the circuit!

No crossing lines, no dead ends — make it a full loop!



MY INNOVATION PLANNING WORKSHOP

Segment 3: Choose Your Invention!

Every inventor picks a problem to solve. Look at the two inventions below and choose the one YOU want to design a circuit for!

💡 Smart Study Lamp – Lights up only when you press the switch. Perfect for reading quietly at night!

🔔 Toy Doorbell – Buzzes when someone opens your door. Know when your sibling is coming!

🔧 Pick one and start sketching your circuit blueprint!



DESIGN OPTION 1: SMART STUDY LAMP

Design a lamp that lights up only when you press the switch.

Your circuit needs:

- 🔋 Battery – gives the energy
- Red wire – outgoing highway
- 💡 LED – the shining star
- Blue wire – return path
- ⚙ Switch – the gatekeeper

Press the switch → close the loop → lamp glows!

Red wire = outgoing path ●
Blue wire = return path ●



DESIGN OPTION 2: TOY DOORBELL

Build a buzzer that sounds when someone opens your door!

Connect your circuit:

● Battery → Switch → Buzzer (red path, outgoing)

● Buzzer → Battery (blue path, return)

When the door opens, it presses the switch — electricity flows and the buzzer sounds!

Try it: Trace the red and blue pathways on your blueprint sheet.

● **Red Path = Outgoing Highway**

● **Blue Path = Return Path**





CONNECT THE CIRCUIT

1. ● Battery to Switch: Clip a jumper wire from the Positive Red terminal of your battery holder to the incoming terminal of your switch.
2. ● Switch to LED: Connect a second wire from the outgoing terminal of your switch to the Long Leg (+) of your LED light bulb.
3. ● LED to Battery: Connect a final wire from the Short Leg (-) of your LED light directly back to the Negative Black terminal of your battery pack.

INSPECTOR'S FIX-IT CHALLENGE!

Be the Electricity Inspector! Something is wrong in these circuits. Can you spot the fault and fix it before the lights go out?



Circuit 1: Battery connected to itself — no loop formed!

Circuit 2: Wire not attached to LED — light won't glow!

Circuit 3: Switch placed in a broken loop — electricity is stuck!

Your job: Find the gap, fix the wire, and close the loop!

FIX THE CIRCUIT – SOLUTIONS!

Great work, Inspectors! Here's how to fix each broken circuit. Spot the gap, fix the wire, and close the loop to get the electricity flowing again!

Close the loop: connect the wire back to the battery so electricity can travel all the way around!

Great job, Inventors! Every fixed circuit lights up the village – keep inventing!



YOUNG INVENTOR SHOWCASE!

Amazing work, Junior Inventors! You have learned how circuits work and built your very own designs. You've earned the T4 Innovation Capstone Badge!

You mastered Complete Loops — no gaps in your circuit highway!

You controlled electricity using Switches like a pro gatekeeper!

Keep inventing, Engineer! The world needs your bright ideas!





THANK YOU, JUNIOR INVENTORS!

Keep Exploring, Keep Inventing!

You've done something amazing today — you designed, built, and debugged real circuits! Every great engineer started just like you. The world needs your ideas!

Keep inventing, Engineer! The future is yours to light up!